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EVEN END SEMESTER EXAMINATION – 2025

Name of the Course: B.Tech
EE

Semester: 8th

Name of the Paper: Wind and
Solar Energy Systems

Paper Code: TEE802

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	CO1, CO2
(a)	Discuss the role of temperature on photo current? Derive the relations for it.	
(b)	How the open circuit voltage is effected by temperature? Find the relations between temperature and open circuit voltage.	
(c)	A family uses 8 kW of power. Direct solar energy is incident on the horizontal surface at an average rate of 200 W per square meter. If 20% of this energy can be converted to useful electrical energy, how large an area is needed to supply 8 kW?	
Q2	(20 marks)	CO3,CO2
(a)	How does the maximum power point tracking effects on the solar power output.	
(b)	Draw equivalent model of solar PV cell and discuss its salient feature.	
(c)	Determine the power in the wind if the wind speed is 20 m/s and blade length is 50 m. assume air density 1,23kg/m ²	
Q3	(20 marks)	CO3,CO4
(a)	What is a solar pond discuss its environmental effects.	
(b)	With a suitable diagram, explain open cycle Ocean Thermal Electric Conversion(OTEC) system.	
(c)	What is single basin tidal power plant? Enumerate its functioning with suitable diagram	
Q4	(20 marks)	
(a)	Explain the working principle of a hybrid wind-solar energy system	
(b)	Draw the block diagram of wind energy conversion system and explain its various components.	

(c)	The efficiency of a solar cell is 20% and its surface area is 4cm^2 . Find the electrical energy generated in one second, by 3000 cells, connected in a solar panel, if 640J of solar energy is incident on one-meter square area, in one second.	
Q5	(20 marks)	
(a)	Differentiate between Non-conventional energy sources and conventional energy sources.	CO5,CO6
(b)	Discuss the advantages and challenges of integrating wind and solar energy into the power grid.	
(c)	what are the major advantages and disadvantages of a wind power plant?	